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資料結構

Data Structure

Lab 2

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Lab02-Ex1.

Find maximum (Finds the maximum value in an array)

Code

```
int findMax(const vector<int>& arr) {
int  stepcount = 0; //初始化 stepcount

int max = arr[0];
  stepcount ++;//變數初始化 1step

  stepcount++;//迴圈初始化 1 step (assignment i = 0)
for (int i = 1; i < arr.size(); i++) {
stepcount++;//迴圈條件判斷(1step)
stepcount++;//陣列存取(1step)

stepcount++;//比較max(1step)
if (arr[i] > max) {
max = arr[i];
}
}
stepcount++;//跳出迴圈(1step)
stepcount++;//return(1step)
return max;
}
//2(initial)+3*n(loop operations)+1 (loop exit) + 1 (return)
//=4+3n
```

Lab02-Ex2. Simple Nested Loops (Prints all unique pairs of elements)

Simple Nested Loops (Prints all unique pairs of elements)

Code

```
void printPairs(const vector<int>& arr) {
int stepcount = 0;//初始化 stepcount
stepcount ++;//loop initialization (1step)
for (int i = 0; i < arr.size(); i++) {
  stepcount ++;//迴圈條件判斷(1step)
  stepcount ++;//陣列存取(1step)
for (int j = i + 1; j < arr.size(); j++) {
stepcount ++;/迴圈條件判斷(1step)
stepcount ++;//陣列存取(1step)
cout << arr[i] << "," << arr[j] << endl;
stepcount ++;//輸出字串(1step)
stepcount ++;//j++(1step)
}
stepcount ++; //跳出迴圈(step)
stepcount ++;//i++(1step)
}
//跳出迴圈(1step)
}
//1(initial)+4*n(outer for loop)*4*n(inner for loop)+1(跳出迴圈)
//=16n^2+2
```

Lab02-Ex3. (題目編號)	
1. Array copying code、2. Array element counter	
Code	
(a)	(b)

```
void copyArray(const vector<int>&
source, vector<int>& dest) {
int stepcount = 0;//初始化 stepcount
stepcounter ++;//初始化迴圈(step1)
for (int i = 0; i < source.size();
i++) {
stepcounter ++;//陣列存取(1step)
dest[i] = source[i];  stepcount =
stepcount+2;
  //2step(陣列存取+assignment)
stepcounter ++;//i++(step1)
}
```

```
int countElement(const vector<int>&
arr, int target) {
int step = 0;//初始化 step
int count = 0;
step++;//變數初始化(1step)
step++;//初始化迴圈(1step)
for (int i = 0; i < arr.size(); i++) {
step ++;//陣列存取(1step)
if (arr[i] == target) {
step ++;//比較(1step)
count++;
}
```

```
stepcounter ++;//跳出迴圈(step1)
}
//1(initial)+4*n(for loop operation)+1(跳出迴圈)
//=4n+2
```

```
step++;//1 step(i++)
}
step ++;//跳出迴圈(1step)
step ++;//return(1step)
return count;
}
//2(initial)+3*n(for loop operation)+2(跳出迴圈+return)
//=3n+4
```

Discussion Section

1. Why don't we count operations like arr.size()?
因為已經存取好變數所以對整體時間不影響。
2. Why do we drop constants in Big-O notation?
因為 Big-O 看的是最大次方，因為最大次方變化比較大就忽略常數。
3. Looking at the examples above:
Sumarray 實際操作次數較多，我覺得 Big-O 在 sumarray 有到 n^2 但 findNumber 只有 n。